Weak forms and problem setup - Three-material steady diffusion-reaction problem on a heterogeneous 2D domain

1 Problem Overview

We consider a parametrized, steady-state **diffusion–reaction** (screened-Poisson) problem on the two-dimensional square domain

$$\Omega = (0, 7)^2 \subset \mathbb{R}^2.$$

The domain is partitioned into three non-overlapping material subdomains,

$$\overline{\Omega} = \overline{\Omega}_1 \cup \overline{\Omega}_2 \cup \overline{\Omega}_3.$$

- **Ω_1 (obstacles, `mat1`):** eleven non-overlapping unit squares placed in a structured chess-board-like pattern (see Figure 1).
- **Ω_2 (source, `mat2`):** the single unit square $[3, 4] \times [3, 4]$ at the centre of Ω.
- **Ω_3 (background, `mat3`):** the connected remainder

$$\Omega \setminus (\overline{\Omega}_1 \cup \overline{\Omega}_2).$$

Each subdomain Ω_i carries independent piecewise-constant material properties: diffusion coefficient $D_i > 0$, reaction coefficient $\Sigma_i \geq 0$, and volumetric source $Q_i \geq 0$, for $i = 1, 2, 3$.

2 Domain Geometry and Mesh

The outer boundary $\partial\Omega$ is the square $[0, 7]^2$. A homogeneous Dirichlet condition $u = 0$ is imposed on the full boundary. Figure 1 shows the subdomain layout.
The domain is discretised by an unstructured triangular mesh generated with `Gmsh` using characteristic lengths

$$h_{\min} = 0.07/4 = 0.0175, \qquad h_{\max} = 0.14/4 = 0.035.$$

Continuous piecewise-linear ($\mathbb{P}_1$) Lagrange elements are employed via `scikit-fem`; Dirichlet conditions are imposed by static condensation.

3 Strong Form

In each subdomain Ω_i, $i = 1, 2, 3$, the governing equation is

$$-D_i \, \Delta u + \Sigma_i \, u = Q_i \qquad \text{in } \Omega_i, \tag{1}$$

subject to the homogeneous Dirichlet boundary condition

$$u = 0 \qquad \text{on } \partial\Omega. \tag{2}$$

Across each internal interface

$$\Gamma_{ij} = \partial\Omega_i \cap \partial\Omega_j,$$

continuity of solution and normal flux is required:

$$[u]_{\Gamma_{ij}} = 0, \qquad [D \, \nabla u \cdot \mathbf{n}]_{\Gamma_{ij}} = 0. \tag{3}$$

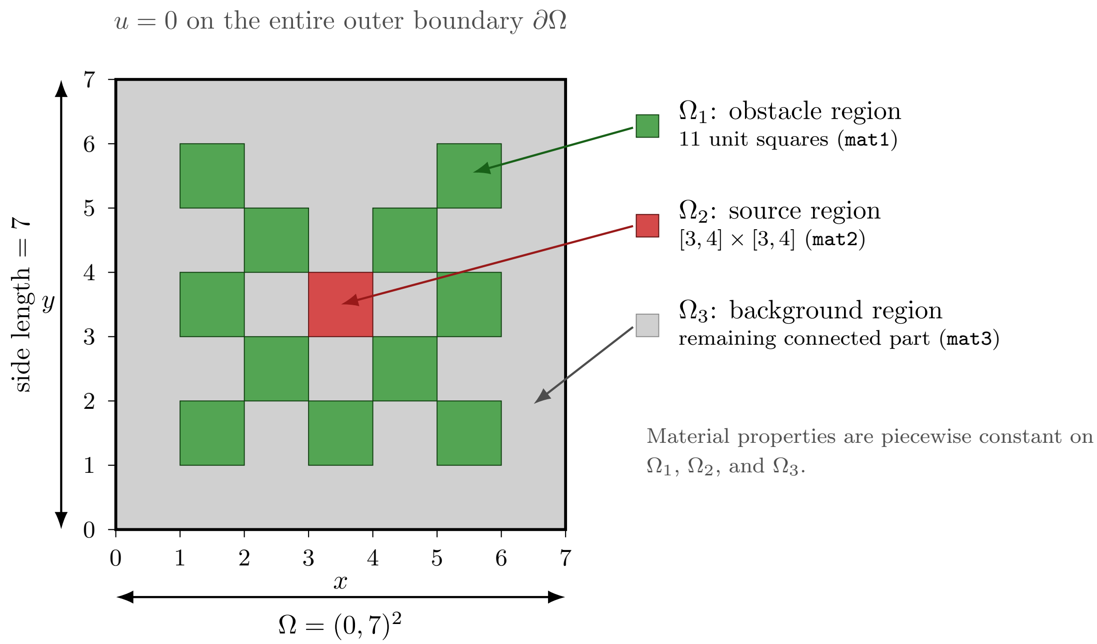

Figure 1: Subdomain layout for the three-material problem on $\Omega = [0,7]^2$. Gray: background Ω_3; green: obstacle region Ω_1 consisting of 11 unit squares; red: central source region $\Omega_2 = [3,4]^2$.

4 Weak Formulation

4.1 Function Space

$$V = H_0^1(\Omega) = \{\, v \in H^1(\Omega) : v|_{\partial\Omega} = 0 \,\}.$$

4.2 Variational Problem

Multiplying (1) by $v \in V$, integrating by parts over each Ω_i, applying the divergence theorem, using (3), and summing over $i = 1, 2, 3$ gives

Find $u \in V$ such that

$$a(u, v; \boldsymbol{\mu}) = \ell(v; \boldsymbol{\mu}) \qquad \forall\, v \in V, \tag{4}$$

where

$$a(u, v; \boldsymbol{\mu}) = \sum_{i=1}^{3} \left[D_i \int_{\Omega_i} \nabla u \cdot \nabla v \, \mathrm{d}\mathbf{x} + \Sigma_i \int_{\Omega_i} uv \, \mathrm{d}\mathbf{x} \right], \tag{5}$$

$$\ell(v; \boldsymbol{\mu}) = \sum_{i=1}^{3} Q_i \int_{\Omega_i} v \, \mathrm{d}\mathbf{x}. \tag{6}$$

Well-posedness

The bilinear form is symmetric and, under $D_i > 0$ and $\Sigma_i \geq 0$, it is coercive:

$$a(v, v; \boldsymbol{\mu}) \geq \min_i (D_i) \, \|\nabla v\|^2_{L^2(\Omega)} > 0 \qquad \forall\, v \in V \setminus \{0\}.$$

By the **Lax–Milgram theorem**, (4) has a unique solution for every admissible $\boldsymbol{\mu}$.

5 Affine Parameter Decomposition

The forms admit the affine decomposition

$$a(u, v; \boldsymbol{\mu}) = \sum_{i=1}^{3} D_i \, a_{\text{diff}}^i(u, v) + \sum_{i=1}^{3} \Sigma_i \, a_{\text{reac}}^i(u, v), \tag{7}$$

$$\ell(v; \boldsymbol{\mu}) = \sum_{i=1}^{3} Q_i \, \ell^i(v), \tag{8}$$

where the parameter-independent component forms are

$$a_{\text{diff}}^i(u, v) = \int_{\Omega_i} \nabla u \cdot \nabla v \, \mathrm{d}\mathbf{x}, \qquad a_{\text{reac}}^i(u, v) = \int_{\Omega_i} uv \, \mathrm{d}\mathbf{x}, \qquad \ell^i(v) = \int_{\Omega_i} v \, \mathrm{d}\mathbf{x}.$$

This yields $Q_a = 6$ affine bilinear terms and $Q_\ell = 3$ affine linear terms.

6 Discrete System

With $\mathbb{P}_1$ basis functions $\{\varphi_j\}$, define the per-subdomain operators

$$[\mathbf{K}_i]_{jk} = \int_{\Omega_i} \nabla \varphi_j \cdot \nabla \varphi_k \, \mathrm{d}\mathbf{x}, \qquad [\mathbf{M}_i]_{jk} = \int_{\Omega_i} \varphi_j \varphi_k \, \mathrm{d}\mathbf{x}, \qquad [\mathbf{f}_i]_j = \int_{\Omega_i} \varphi_j \, \mathrm{d}\mathbf{x}. \tag{9}$$

The global assembled linear system reads

$$\underbrace{\Big(D_1\mathbf{K}_1 + D_2\mathbf{K}_2 + D_3\mathbf{K}_3 + \Sigma_1\mathbf{M}_1 + \Sigma_2\mathbf{M}_2 + \Sigma_3\mathbf{M}_3\Big)}_{\mathbf{A}(\boldsymbol{\mu})} \mathbf{u} = \underbrace{Q_1\mathbf{f}_1 + Q_2\mathbf{f}_2 + Q_3\mathbf{f}_3}_{\mathbf{b}(\boldsymbol{\mu})}. \tag{10}$$

7 Parameter Space

$$\boldsymbol{\mu} = (D_1, D_2, D_3, \Sigma_1, \Sigma_2, \Sigma_3, Q_1, Q_2, Q_3) \in \mathcal{P} \subset \mathbb{R}^9.$$

Parameter	**Physical meaning**	**Zone**	**Range / value**
D_1	Diffusion coeff.	Ω_1 (obstacles)	$[0.1,\ 1.0]$
D_2	Diffusion coeff.	Ω_2 (source)	$[0.1,\ 1.0]$
D_3	Diffusion coeff.	Ω_3 (background)	$[0.1,\ 1.0]$
Σ_1	Reaction coeff.	Ω_1 (obstacles)	$[0.0,\ 4.0]$
Σ_2	Reaction coeff.	Ω_2 (source)	2.0 (fixed)
Σ_3	Reaction coeff.	Ω_3 (background)	$[0.0,\ 6.0]$
Q_1	Volumetric source	Ω_1 (obstacles)	0.0 (fixed)
Q_2	Volumetric source	Ω_2 (source)	10.0 (fixed)
Q_3	Volumetric source	Ω_3 (background)	30.0 (fixed)

Table 1: Admissible parameter ranges. Five parameters $(D_1, D_2, D_3, \Sigma_1, \Sigma_3)$ are active; the remaining four are fixed. Sobol quasi-random sequences are used, with $N_{\text{snap}} = 32$ training samples and 32 test samples.